Strehl ratio of a Gaussian beam

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We discuss the Strehl ratio of systems with a Gaussian pupil and determine the range of validity of its approximate expression based on the aberration variance. The results given are equally applicable to propagation of Gaussian beams. The uniform and weakly truncated pupils are considered as limiting cases of a Gaussian pupil. We show that the approximate expression for Strehl ratio in terms of the aberration variance yields a good estimate of the true value for a strongly truncated pupil but a much smaller value for a weakly truncated pupil. © 2005 Optical Society of America

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1. INTRODUCTION

A simple but important measure of the quality of an image formed by an imaging system is its Strehl ratio, representing the ratio of the central irradiance of its point-spread function with and without aberration.[1–3] For small aberrations, a reasonable estimate of its value can be obtained from a knowledge of the variance of the aberration across its amplitude-weighted pupil.[4–7] We discuss the Strehl ratio of optical systems with Gaussian apodization or Gaussian pupils, i.e., those with a Gaussian amplitude across the wavefront at their exit pupils. The discussion is equally applicable to imaging systems with a Gaussian transmission (obtained, for example, by placing a Gaussian filter at its exit pupil) as well as laser transmitters in which the laser beam has a Gaussian distribution at its exit pupil.

For uniform pupils, it has been shown that the Strehl ratio for primary aberrations can be approximated by a simple expression in terms of their aberration variance across the pupil.[8] While a similar expression holds for the Strehl ratio of a Gaussian pupil, where the aberration variance is calculated across the apodized or amplitude-weighted pupil, its range of validity has not been explored. Similarly although exact closed-form expressions have been given for some primary aberrations (without any balancing), no comparison with the approximate results was made.[9] We show that a balanced aberration does yield a higher Strehl ratio, as expected, but also that a higher Strehl ratio is obtained for large coma and astigmatism without any balancing. We also show that the approximate value of Strehl ratio obtained from its expression in terms of the aberration variance yields a good estimate of the true value for a strongly truncated pupil, but a much smaller value for a weakly truncated pupil.

2. GAUSSIAN PUPIL

A pupil is referred to as a Gaussian pupil if the amplitude variation across it has the form of a Gaussian. It can be obtained in two different ways. In imaging applications, the wave incident on the exit pupil has a uniform amplitude, but its transmission varies as a Gaussian. The pupil is said to be apodized. In applications of Gaussian beam propagation, the wave incident on the pupil has a Gaussian amplitude, but its transmission is uniform. In either case, the transmitted amplitude immediately to the right of a circular pupil of radius a and area $S_p=\pi a^2$ can be written

$$A(\rho)=A_0\exp(-\gamma\rho^2), \qquad 0\leqslant\rho\leqslant 1, \tag{1}$$

where ρ is the radial distance of a point in the pupil from its center normalized by its radius a, A_0 is a constant (with A_0^2 having dimensions of $\mathrm{W/m^2}$), and

$$\gamma=(a/\omega)^2 \tag{2}$$

is a truncation parameter of the pupil with ω as the Gaussian radius representing the radial distance at which the amplitude decreases to $1/e$ of its value at the center. The case of a uniform pupil is obtained by letting $\gamma\to 0$. A large value of γ or $\omega\ll a$ represents a narrow Gaussian, or a weakly truncated pupil.

The total power incident on the pupil plane is given by

$$P_{inc}=2S_pA_0^2\int_0^\infty \exp(-2\gamma\rho^2)\rho\mathrm{d}\rho=\frac{S_pA_0^2}{2\gamma}. \tag{3}$$

Similarly the total power transmitted by the pupil is given by

$$P_p=2S_pA_0^2\int_0^1 \exp(-2\gamma\rho^2)\rho\mathrm{d}\rho=(S_p/2\gamma)A_0^2[1-\exp(-2\gamma)]. \tag{4}$$

Hence the fraction of the power transmitted by the pupil is given by

$$P_{trans}=P_p/P_{inc}=1-\exp(-2\gamma). \tag{5}$$

The fractional power transmitted by the pupil as a function of $\sqrt{\gamma}$ or a/ω is illustrated in Fig. 1 by the curve labeled "Beam." Its value is 86.47% when $\gamma=1$, 98.89% when $\sqrt{\gamma}=1.5$, 99.97% when $\sqrt{\gamma}=2$, and practically 100% when $\sqrt{\gamma}=3$. Thus the fractional power $\exp(-2\gamma)$ lying outside the pupil is negligibly small when $\sqrt{\gamma}\geqslant 2$.

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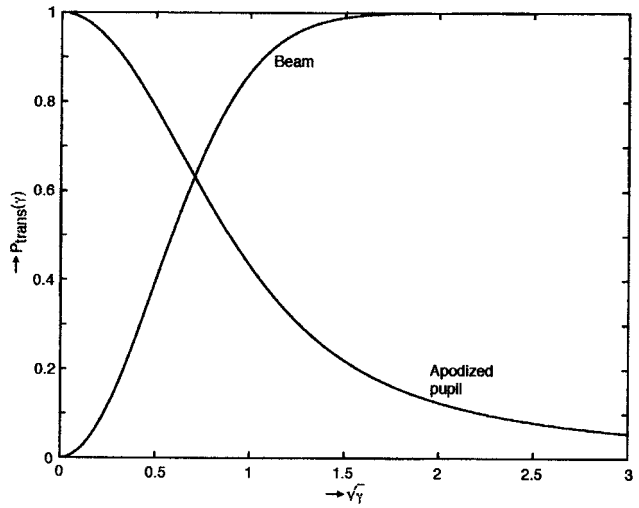


Fig. 1. Fractional power P_{trans} transmitted by the exit pupil as a function of $\sqrt{\gamma}=a/\omega$. The curve labeled "Beam" is for a Gaussian beam incident on a pupil that has uniform transmission, and that labeled "Apodized pupil" is for uniform illumination of a pupil with a Gaussian transmission.

For a uniformly illuminated pupil that transmits in the form of a Gaussian, e.g., by placing a Gaussian amplitude filter, the power incident on it is given by

$$P_{inc}=A_0^2 S_p. \tag{6}$$

The power transmitted by the pupil is again given by Eq. (4). Hence the fractional transmitted power in this case is given by

$$P_{trans}=P_p/P_{inc}=\frac{1-\exp(-2\gamma)}{2\gamma}. \tag{7}$$

It decreases as γ increases, as illustrated in Fig. 1 by the curve labeled "Apodized pupil." Its value for $\sqrt{\gamma}=1$, 2, and 3 is 43.23%, 12.50%, and 5.56%, respectively. For a Gaussian beam as well as a Gaussian apodized pupil, the irradiance A_0^2 at its center for a total power P_{trans} transmitted by the pupil is given by

$$A_0^2=\frac{2\gamma P_{trans}}{S_p[1-\exp(-2\gamma)]}. \tag{8}$$

The purpose of apodization using an amplitude filter is to reduce the secondary maxima of the point-spread function of a system. Although the secondary maxima decrease as γ increases, a typical value considered is $\gamma=1$ to avoid excessive transmission loss with increasing γ. The low-power laser beams, e.g., He-Ne beams, are generally weakly truncated. They are represented in this paper by $a=3\omega$ or $\sqrt{\gamma}=3$. Although 99.97% of the power of a beam is transmitted by a pupil with $a=2\omega$, or $\sqrt{\gamma}=2$, the actual standard deviation of an aberration can be quite different from its value obtained under the weakly truncated assumption.[2,10]

3. STREHL RATIO AND ABERRATION VARIANCE

The irradiance distribution of the image of a point object for an aberration-free system is maximum at the Gaussian image point. Similarly, a focused beam yields maximum irradiance at its focal point when it is aberration free. This value of irradiance is reduced when an aberration is present. Let R be the distance of the image point or the focal point from the pupil. By definition, the Strehl ratio of the system or a beam is the ratio of the central irradiances at a certain distance z with and without aberration. Hence the Strehl ratio may be written[2]

$$S=\left\{\frac{\gamma}{\pi[1-\exp(-\gamma)]}\right\}^2\left|\int_0^1\int_0^{2\pi}\exp(-\gamma\rho^2)\times\exp[i\Phi(\rho,\theta)]\rho\,\mathrm{d}\rho\,\mathrm{d}\theta\right|^2, \tag{9}$$

where $\Phi(\rho,\theta)$ is the phase aberration at a point (ρ,θ) on the pupil with respect to a reference sphere of radius of curvature R centered at the Gaussian image point or the focal point. Its value, which is less than one, represents the effect of aberration on the central irradiance.

For small aberrations, approximate expressions for the Strehl ratio may be written[2,3,8,9]

$$S_1 > \simeq (1-\sigma_\Phi^2/2)^2, \tag{10a}$$

$$S_2 \simeq 1-\sigma_\Phi^2, \tag{10b}$$

$$S_3 \simeq \exp(-\sigma_\Phi^2), \tag{10c}$$

where

$$\sigma_\Phi^2=\langle\Phi^2\rangle-\langle\Phi\rangle^2 \tag{11}$$

is the variance of the phase aberration across the Gaussian-amplitude weighted pupil. The mean and the mean square values of the aberration are given by

$$\langle\Phi^n\rangle=\int_0^1\int_0^{2\pi}A(\rho)[\Phi(\rho,\theta)]^n\rho\,\mathrm{d}\rho\,\mathrm{d}\theta\Big/\int_0^1\int_0^{2\pi}A(\rho)\rho\,\mathrm{d}\rho\,\mathrm{d}\theta$$
$$=\frac{\gamma}{\pi[1-\exp(-\gamma)]}\int_0^1\int_0^{2\pi}\exp(-\gamma\rho^2)[\Phi(\rho,\theta)]^n\rho\,\mathrm{d}\rho\,\mathrm{d}\theta \tag{12}$$

with $n=1$ and 2, respectively. Note that $S_1>S_2$ by $\sigma_\Phi^4/4$ and $S_3>S_1$ by $\sigma_\Phi^4/4$. For a pupil illuminated with uniform amplitude, both S_1 and S_2 underestimate the Strehl ratio, but S_3 yields values within 10% of the true value for $S\geqslant 0.3$.[8]

4. WEAKLY TRUNCATED GAUSSIAN PUPILS

When the Gaussian illumination is very narrow compared with the radius of the pupil, i.e., when $\omega\ll a$ or γ is very large, it is said to be weakly truncated. For such an illumination, the pupil may be assumed to be infinitely wide with the result that a Gaussian beam exiting from the pupil remains Gaussian as it propagates.[2,11,12] Moreover $\exp(-2\gamma)$ in Eq. (8) may be neglected compared with unity, and Eq. (1) for the pupil amplitude may be written

$$A(\rho) = (2\gamma P_{trans}/S_p)^{1/2} \exp(-\gamma\rho^2). \quad (13)$$

Similarly, letting the upper limit on the radial integration approach infinity, Eq. (9) for the Strehl ratio and Eq. (12) for the mean and mean square values of the aberration may be written

$$S = \left(\frac{\gamma}{\pi}\right)^2 \left| \int_0^\infty \int_0^{2\pi} \exp(-\gamma\rho^2)\exp[i\Phi(\rho,\theta)]\rho \mathrm{d}\rho \mathrm{d}\theta \right|^2, \quad (14)$$

$$\langle \Phi^n \rangle = \frac{\gamma}{\pi} \int_0^\infty \int_0^{2\pi} \exp(-\gamma\rho^2)[\Phi(\rho,\theta)]^n \rho \mathrm{d}\rho \mathrm{d}\theta, \quad (15)$$

respectively.

Table 1 lists the Seidel or primary aberrations and their standard deviations for increasing values of γ. It is evident that the standard deviation of an aberration decreases as γ increases. This is because while an aberration increases as ρ increases, the amplitude decreases more and more rapidly as γ increases and reduces the effect of the aberration more and more compared with that for a uniform pupil. Accordingly for a given small amount of aberration A_i, the Strehl ratio for a Gaussian pupil is higher than that for a uniform pupil. Similarly, the aberration tolerance for a given Strehl ratio is higher for a Gaussian pupil.

5. DEPTH OF FOCUS

Letting $\Phi(\rho,\theta) = B_d\rho^2$ in Eq. (9), where B_d is the coefficient of the defocus phase aberration representing its peak value, we obtain the Strehl ratio of a defocused but otherwise aberration-free beam:

$$S = \left[\frac{\gamma}{1-\exp(-\gamma)}\right]^2 \left| \int_0^1 \exp[-(\gamma - iB_d)x]\mathrm{d}x \right|^2$$
$$= \left[\frac{\gamma}{1-\exp(-\gamma)}\right]^2 \frac{1}{\gamma^2 + B_d^2}[1 - \exp(-2\gamma) - 2\exp(-\gamma)\cos B_d]. \quad (16)$$

It is evident that S is independent of the sign of B_d. Multiplying the Strehl ratio by the focal-point irradiance $(P_p S_p/\lambda^2 z^2)[\tanh(\gamma/2)/(\gamma/2)]$ of the beam when it is focused at a distance z yields its axial irradiance.[13] For a uniform or a weakly truncated Gaussian beam, Eq. (16) reduces to

$$S = \left(\frac{\sin B_d/2}{B_d/2}\right)^2, \quad (17)$$

or

$$S = \frac{1}{1 + (B_d/\gamma)^2}, \quad (18)$$

respectively.

The defocus coefficient for a beam focused at a distance R but observed at a distance z is given by

$$B_d(z) = \pi N\left(\frac{R}{z} - 1\right), \quad (19)$$

where $N = a^2/\lambda R$ is the Fresnel number representing the number of Fresnel zones in the pupil as observed from the focus. When N is very small, as in laser transmitters, B_d is relatively small even when z is much different from R, thus yielding a large depth of focus. Hence defocus in this case affects the central irradiance in two ways. It reduces the central irradiance as a result of the defocus aberration but increases or decreases it as a result of the inverse-square-law dependence on z for $z < R$ or $z > R$, respectively. Hence for small Fresnel numbers, the depth of focus depends not only on the Strehl ratio but also on the $(R/z)^2$ factor. Accordingly, the large depth of focus is asymmetrical about the focal point.

For large Fresnel numbers, as in imaging systems, B_d is large even for very small differences between z and R, thus yielding a small depth of focus. Accordingly, the small defocus aberration tolerance dictates that z be approximately equal to R. Hence, Eq. (19) may be written

$$B_d(z) = \pi N(1 - z/R). \quad (20)$$

In this case, the depth of focus or a longitudinal defocus $\Delta = z - R$ is related to the defocus coefficient according to

$$\Delta = -8B_d F^2, \quad (21)$$

where $F = R/2a$ is the focal ratio (or the f-number) of the focused beam cone and B_d is now the coefficient of the defocus wave aberration [obtained by multiplying the phase aberration coefficient B_d in Eq. (19) by $\lambda/2\pi$]. The depth of focus is symmetrical about the focal point, since the defocus coefficients for $\pm\Delta$ have equal magnitudes though opposite signs.

Table 1. Primary Aberrations and Their Standard Deviations for Optical Systems with Gaussian Pupils[a]

Seidel Aberration	$\sigma_\Phi(\gamma=0)$	$\sigma_\Phi(\gamma=1)$	$\sigma_\Phi(\sqrt{\gamma}=2)$	$\sigma_\Phi(\sqrt{\gamma}\geq 3)$
Spherical, $A_s\rho^4$	$2A_s/3\sqrt{5} = A_s/3.35$	$A_s/3.67$	$A_s/6.20$	$2\sqrt{5}A_s/\gamma^2$
Coma, $A_c\rho^3\cos\theta$	$A_c/2\sqrt{2} = A_c/2.83$	$A_c/3.33$	$A_c/6.08$	$\sqrt{3}A_c/\gamma^{3/2}$
Astigmatism, $A_a\rho^2\cos^2\theta$	$A_a/4$	$A_a/4.40$	$A_a/6.59$	$A_a/\sqrt{2}\gamma$
Defocus, $B_d\rho^2$	$B_d/2\sqrt{3} = B_d/3.46$	$B_d/3.55$	$B_d/4.79$	B_d/γ
Tilt, $B_t\rho\cos\theta$	$B_t/2$	$B_t/2.19$	$B_t/2.94$	$B_t/\sqrt{2\gamma}$

[a]For comparison, the results for a uniform pupil, i.e., $\gamma=0$, are also given.

Table 2. Balanced Primary Aberrations

Balanced Aberration	$\Phi(\rho,\theta;\gamma=0)$	$\Phi(\rho,\theta;\gamma=1)$	$\Phi(\rho,\theta;\sqrt{\gamma}=2)$	$\Phi(\rho,\theta;\sqrt{\gamma}\geq 3)$
Spherical	$A_s(\rho^4-\rho^2)$	$A_s(\rho^4-0.933\rho^2)$	$A_s(\rho^4-0.728\rho^2)$	$A_s(\rho^4-4\rho^2/\gamma)$
Coma	$A_c(\rho^3-2\rho/3)\cos\theta$	$A_c(\rho^3-0.608\rho)\cos\theta$	$A_c(\rho^3-0.419\rho)\cos\theta$	$A_c(\rho^3-2\rho/\gamma)\cos\theta$
Astigmatism	$A_a\rho^2(\cos^2\theta-1/2)$	$A_a\rho^2(\cos^2\theta-1/2)$	$A_a\rho^2(\cos^2\theta-1/2)$	$A_a\rho^2(\cos^2\theta-1/2)$

6. BALANCED ABERRATIONS

Since the Strehl ratio depends on the aberration variance, we balance a given aberration with lower order aberrations to minimize its variance. Thus we balance spherical aberration and astigmatism with defocus aberration, and coma with tilt aberration. We calculate the variance of the balanced aberration and equate its derivative with respect to the balancing aberration to zero to determine its optimum value to minimize the variance. The balanced primary aberrations thus obtained are listed in Table 2.[10,14–19] These balanced aberrations can be identified with Zernike circle polynomials when $\gamma=0$ and Zernike–Gauss polynomials when $\gamma\neq 0$.[2,10,15–19] Accordingly they may be referred to as the orthogonal or Zernike aberrations. Their standard deviations are given in Table 3. The factor by which the standard deviation of a primary aberration is reduced by balancing it with another is listed in Table 4. The diffraction focus representing the point of maximum irradiance for a small aberration is listed in Table 5. We note that, although aberration balancing in the case of a uniform pupil reduces the standard deviation of spherical aberration and coma by factors of 4 and 3, respectively, the reduction in the case of astigmatism is a factor of only 1.225. For a Gaussian pupil, the trend is similar but the reduction factors are smaller for spherical aberration and coma and larger for astigmatism. For a Gaussian beam with $\gamma=1$, they are 3.74, 2.64, and 1.27 for spherical aberration, coma, and astigmatism, respectively. The corresponding numbers for a weakly truncated beam are $\sqrt{5}$, $\sqrt{3}$, and $\sqrt{2}$. The amount of balancing aberration decreases as γ increases in the case of spherical aberration and coma, but it does not change in the case of astigmatism. For example, in the case of spherical aberration, the amount of balancing defocus for a weakly truncated Gaussian beam is $(4/\gamma)$ times the corresponding amount for a uniform beam. Similarly, in the case of coma, the balancing tilt is $(3/\gamma)$ times the amount for a uniform beam. The diffraction focus moves closer to the focal point as γ increases in the case of spherical aberration and coma, but does not change in the case of astigmatism. Compared with the peak value, the standard deviation of an aberration in a weakly truncated beam is smaller by a factor of $\gamma^2/2$, $\gamma^{3/2}$, and 2γ in the case of spherical aberration, coma, and astigmatism, respectively.

For spherical aberration $A_s\rho^4$ balanced with defocus $B_d\rho^2$, for coma $A_c\rho^3\cos\theta$ balanced with tilt $B_t\rho\cos\theta$, and for astigmatism $A_a\rho^2\cos^2\theta$ balanced with defocus $B_d\rho^2$, Eq. (9) reduces to

$$S=\left[\frac{\gamma}{1-\exp(-\gamma)}\right]^2\left|\int_0^1\exp(-\gamma x)f(x)\mathrm{d}x\right|^2, \qquad (22)$$

where

Table 3. Standard Deviation of Balanced Primary Aberrations

Balanced Aberration	$\sigma_\Phi(\gamma=0)$	$\sigma_\Phi(\gamma=1)$	$\sigma_\Phi(\sqrt{\gamma}=2)$	$\sigma_\Phi(\sqrt{\gamma}\geq 3)$
Spherical	$A_s/6\sqrt{5}=A_s/13.42$	$A_s/13.71$	$A_s/18.29$	$2A_s/\gamma^2$
Coma	$A_c/6\sqrt{2}=A_c/8.49$	$A_c/8.80$	$A_c/12.21$	$A_c/\gamma^{3/2}$
Astigmatism	$A_a/2\sqrt{6}=A_a/4.90$	$A_a/5.61$	$A_a/9.08$	$A_a/2\gamma$

Table 4. Factor by Which the Standard Deviation of a Seidel Aberration Is Reduced When Optimally Balanced with Other Aberrations

	Reduction Factor			
Balanced Aberration	Uniform ($\gamma=0$)	Gaussian ($\gamma=1$)	Gaussian ($\sqrt{\gamma}=2$)	Weakly Truncated Gaussian ($\sqrt{\gamma}\geq 3$)
Spherical	4	3.74	2.95	$\sqrt{5}=2.24$
Coma	3	2.64	2.01	$\sqrt{3}=1.73$
Astigmatism	1.225	1.27	1.38	$\sqrt{2}=1.41$

$$f(x)=\begin{cases}\exp[i(A_sx^2+B_dx)] & \text{Spherical + defocus}\\ J_0(A_cx^{3/2}+B_tx^{1/2}) & \text{Coma + tilt}\\ \exp[i(0.5A_a+B_d)x]J_0(0.5A_ax) & \text{Astigmatism + defocus}\end{cases}. \qquad (23)$$

The coefficient of a balancing aberration for different values of γ is given in Table 2. For example, the defocus aberration that balances spherical aberration is given by $B_d/A_s=-1$, -0.933, -0.728, and $-4/\gamma$ when $\sqrt{\gamma}=0$, 1, 2, and ≥ 3, respectively. Similarly the tilt aberration that balances coma for these values of γ is given by $B_t/A_c=-(2/3)$, -0.608, -0.419, and $-2/\gamma$, respectively. The defocus coefficient given by $B_d=-A_a/2$ to balance astigmatism is independent of γ.

The Strehl ratio can be expressed in terms of the standard deviation σ_Φ of an aberration instead of its coefficient A_i by utilizing their relationship given in Tables 1 and 3. It has been shown that for a uniform pupil, a closed-form expression for the Strehl ratio is obtained for spherical aberration and astigmatism with and without aberration balancing.[2,4,5] The corresponding closed-form expressions for a weakly truncated Gaussian pupil, except for balanced spherical aberration, are given in Table

Table 5. Diffraction Focus

Balanced Aberration	Diffraction Focus: Uniform ($\gamma=0$)	Gaussian ($\gamma=1$)	Gaussian ($\sqrt{\gamma}=2$)	Weakly Truncated Gaussian ($\sqrt{\gamma}\geq 3$)
Spherical	$(0,0,8F^2A_s)$	$(0,0,7.46F^2A_s)$	$(0,0,5.82F^2A_s)$	$(0,0,32F^2A_s/\gamma)$
Coma	$(4FA_c/3,0,0)$	$(1.22FA_c,0,0)$	$(0.84FA_c,0,0)$	$(4FA_c/\gamma,0,0)$
Astigmatism	$(0,0,4F^2A_a)$	$(0,0,4F^2A_a)$	$(0,0,4F^2A_a)$	$(0,0,4F^2A_a)$

Table 6. Strehl Ratio Expressions for Uniform and Weakly Truncated Gaussian Beams in Terms of the Standard Deviation σ_i of an Aberration[a]

Aberration	$S(\gamma=0)$	$S(\sqrt{\gamma}\geq 3)$
Spherical, $A_s\rho^4$	$(1/b)\{C^2(\sqrt{b})+S^2(\sqrt{b})\}$	$(\pi^2/b')\{[1/2-S(1/\sqrt{b'})]^2+[1/2-C(1/\sqrt{b'})]^2\}$
Balanced spherical, $A_s\rho^4+B_d\rho^2$	$(1/b)\{C^2(\sqrt{b})+S^2(\sqrt{b})\}$	Eq. (22) with $B_d=-(4/\gamma)A_s$ and $A_s=\gamma^2\sigma_s/2$
Coma, $A_c\rho^3\cos\theta$	$[\int_0^1 J_0(2\sqrt{2}\sigma_c x^{3/2})\mathrm{d}x]^2$	Eq. (22) with $B_t=0$ and $A_c=\gamma^{3/2}\sigma_c/\sqrt{3}$
Balanced coma, $(A_c\rho^3+B_t\rho)\cos\theta$	$[\int_0^1 J_0[6\sqrt{2}\sigma_{bc}(x^{3/2}-2x^{1/2}/3)]\mathrm{d}x]^2$	Eq. (22) with $B_t=-(2/\gamma)A_c$ and $A_c=\gamma^{3/2}\sigma_{bc}$
Astigmatism, $A_a\rho^2\cos^2\theta$	$J_0^2(2\sigma_a)+J_1^2(2\sigma_a)$	$[1+(A_a/\gamma)^2]^{-1/2}=[1+2\sigma_a^2]^{-1/2}$
Balanced astigmatism, $A_a\rho^2(\cos^2\theta-1/2)$	$2/3\sigma_{ba}^2[\sum_{k=0}^{\infty} J_{2k+1}(\sqrt{6}\sigma_{ba})]^2$	$[1+(A_a/2\gamma)^2]^{-1}=1/(1+\sigma_{ba}^2)$
Defocus, $B_d\rho^2$	$(\sin\sqrt{3}\sigma_d/\sqrt{3}\sigma_d)^2$	$[1+(B_d/\gamma)^2]^{-1}=1/(1+\sigma_d^2)$

[a] $b=3\sqrt{5}\sigma_s/\pi$, $b'=\pi\sigma_s/\sqrt{5}$, $C(b)=\int_0^b\cos(\pi x^2/2)\mathrm{d}x$, $S(b)=\int_0^b\sin(\pi x^2/2)\mathrm{d}x$.

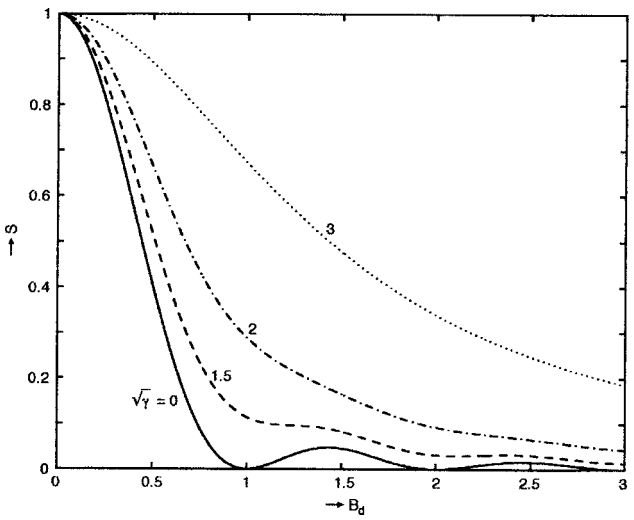


Fig. 2. Strehl ratio of a Gaussian beam as a function of the defocus wave aberration coefficient B_d in units of wavelength λ, showing how it increases as γ increases. For a certain value of the Strehl ratio, the value of B_d is obtained from this figure which, in turn, is used to obtain the corresponding value of the depth of focus from Eq. (21).

6. Integral expressions are given when a closed-form expression is not possible. For comparison and completeness, the corresponding expressions for a uniform pupil are also given in the table. In particular, the Strehl ratio for a given value of the standard deviation is the same for both spherical and balanced spherical aberrations.[2,8] Of course, for the same value of standard deviation, the aberration coefficient A_s for spherical aberration is one fourth of that for the balanced spherical aberration. As a reminder, we add that minimum aberration variance yields maximum Strehl ratio only for small aberrations. For large aberrations, there is no simple relationship between the Strehl ratio and the aberration variance.[2,4,5,20]

In the case of a weakly truncated Gaussian beam, we can immediately draw some conclusions because of the simplicity of the closed-form expressions for the Strehl ratio for astigmatism, balanced astigmatism, and defocus. For a given value of σ, i.e., for $\sigma_a=\sigma_{ba}$, the Strehl ratio for balanced astigmatism is smaller than that for astigmatism. This does not mean that aberration balancing for minimum variance degrades the Strehl ratio in this case, since the corresponding aberration coefficient A_a for balanced astigmatism is larger by a factor of $\sqrt{2}$ compared with that for astigmatism. This may also be seen from the Strehl ratio expression in terms of A_a. For a given value of A_a, the Strehl ratio for balanced astigmatism is larger than that for astigmatism for $A_a\leq 4\lambda$ (see Fig. 6 below). For a given value of σ, the Strehl ratio for defocus is exactly the same as for balanced astigmatism, and therefore smaller than that for astigmatism. However, if $A_a=B_d$, then the Strehl ratio for balanced astigmatism is higher than that for defocus.

7. NUMERICAL RESULTS

Figure 2 shows how the Strehl ratio varies with the defocus aberration B_d in units of wavelength λ for various values of γ. It is evident that for a given value of S, the de-

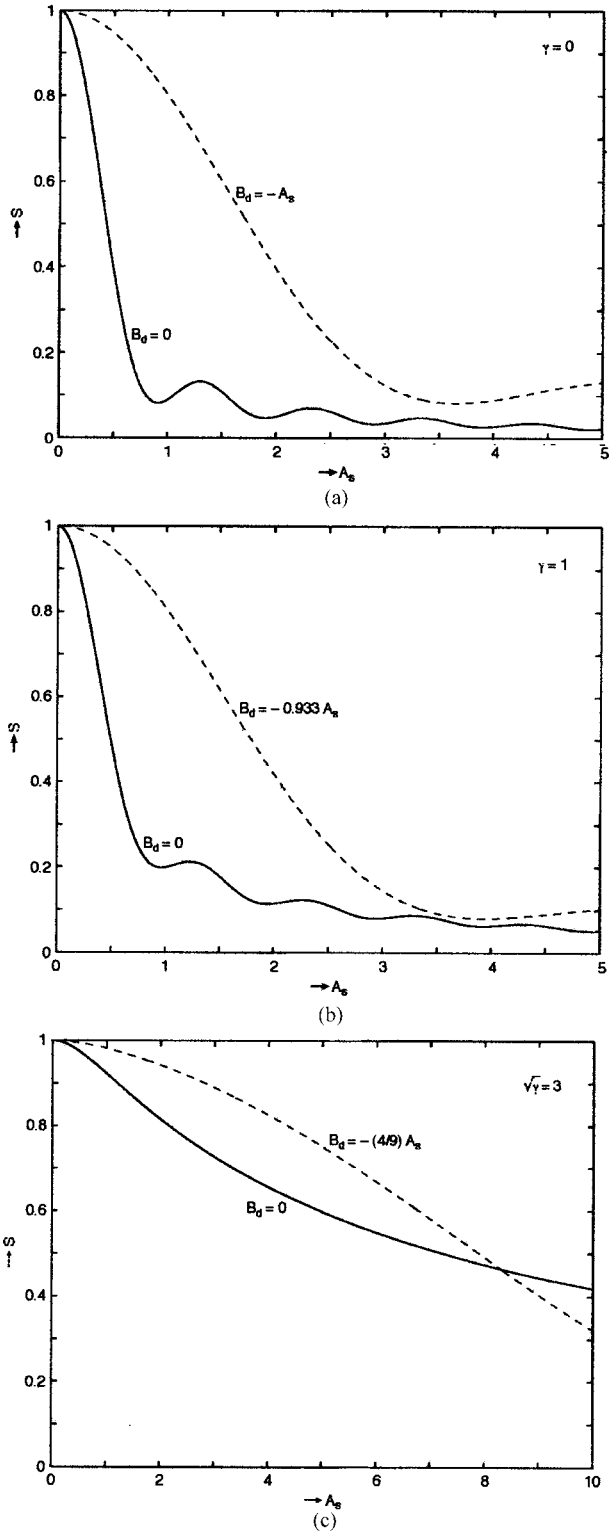


Fig. 3. Strehl ratio of a beam aberrated by spherical aberration A_s in units of wavelength λ and balanced by defocus aberration B_d. (a) Uniform beam ($\gamma=0$). (b) Gaussian beam with $\gamma=1$. (c) Weakly truncated Gaussian beam with $\sqrt{\gamma}=3$.

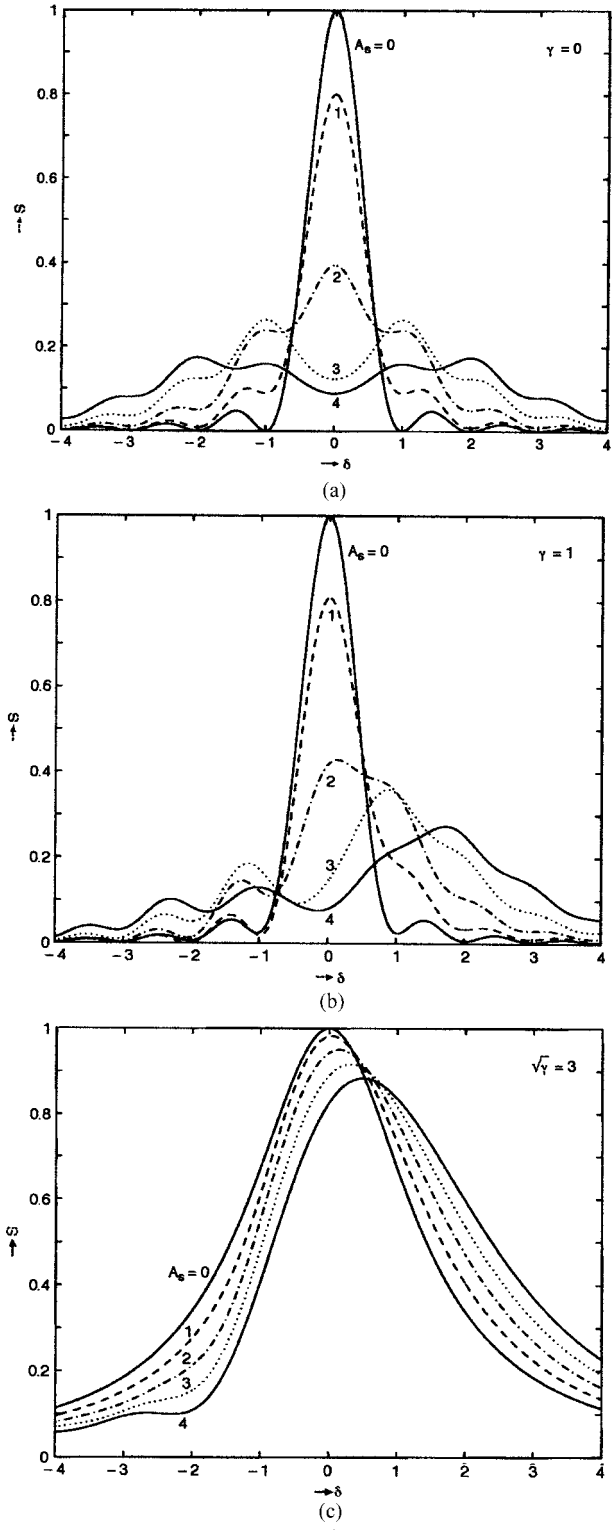


Fig. 4. Strehl ratio for balanced spherical aberration as a function of defocus aberration δ, which represents the deviation from optimum balancing of spherical aberration for minimum variance. Both A_s and δ are in units of wavelength λ. (a) Uniform beam ($\gamma=0$) and $\delta=B_d+A_s$. (b) Gaussian beam with $\gamma=1$ and $\delta=B_d+0.933A_s$. (c) Gaussian beam with $\sqrt{\gamma}=3$ and $\delta=B_d+(4/\gamma)A_s$.

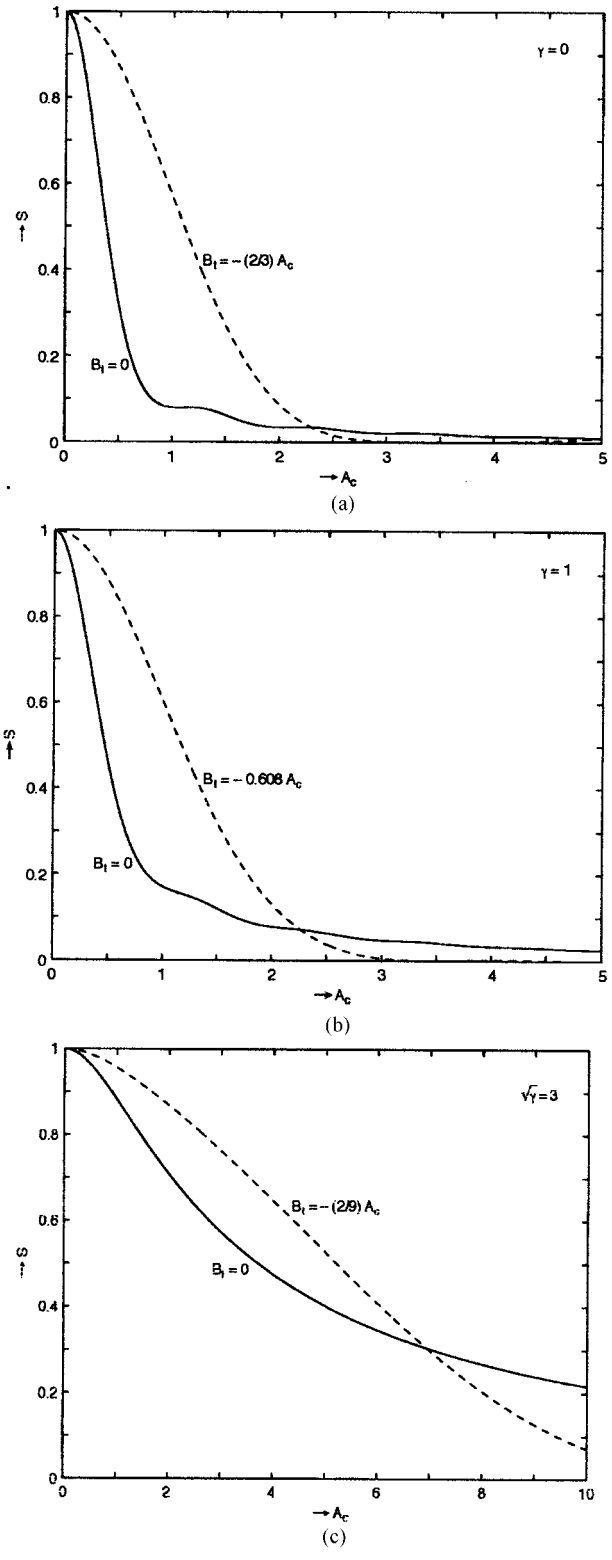


Fig. 5. Strehl ratio of beam aberrated by coma A_c in units of wavelength λ. (a) Uniform beam ($\gamma=0$). (b) Gaussian beam with $\gamma=1$. (c) Weakly truncated Gaussian beam with $\sqrt{\gamma}=3$.

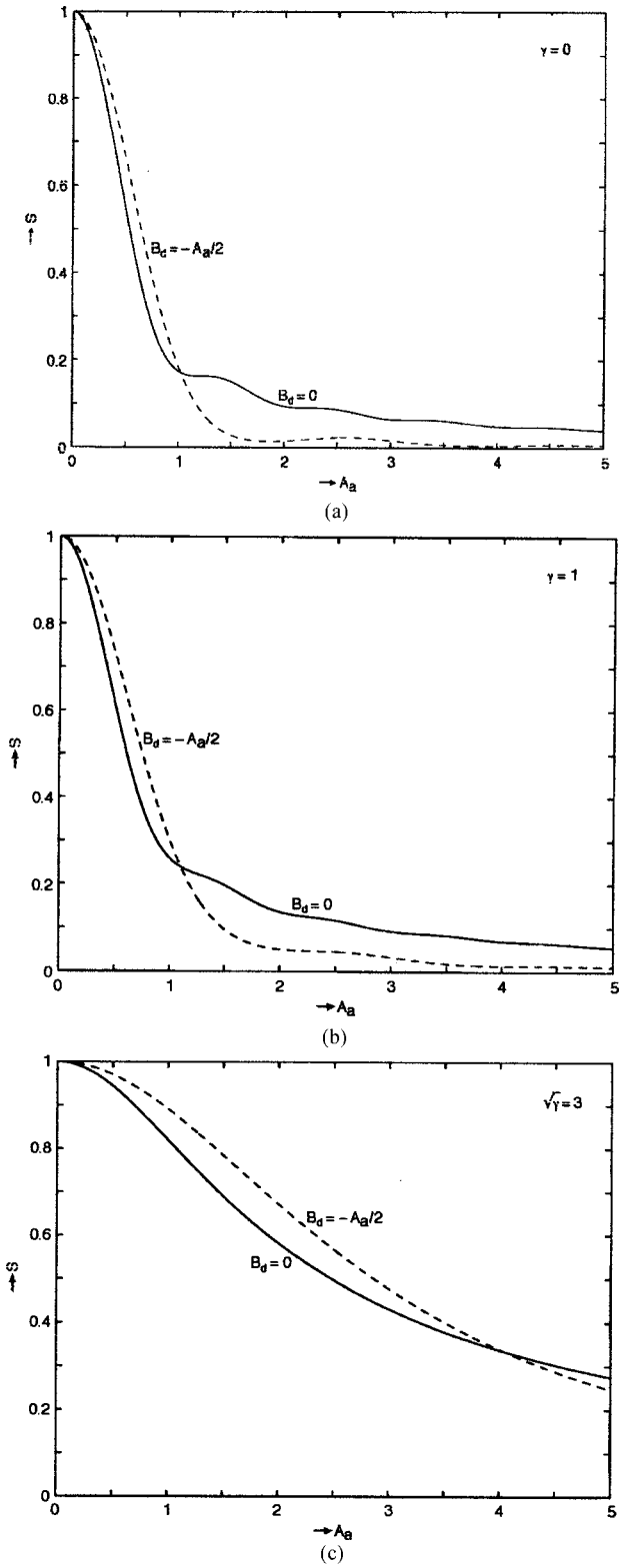


Fig. 6. Strehl ratio of beam aberrated by astigmatism A_a in units of wavelength λ. (a) Uniform beam ($\gamma=0$). (b) Gaussian beam with $\gamma=1$. (c) Weakly truncated Gaussian beam with $\sqrt{\gamma}=3$.

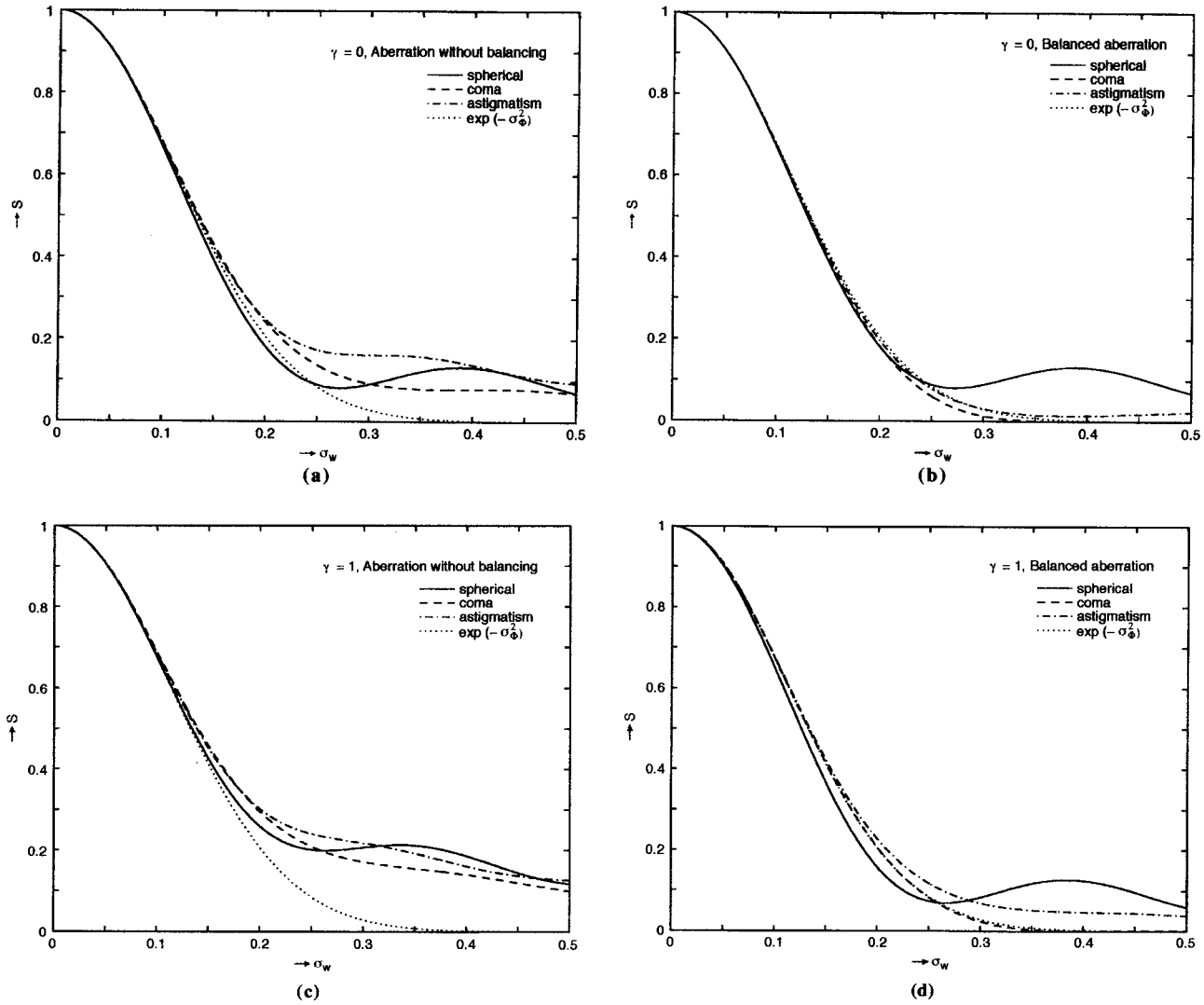


Fig. 7. Strehl ratio of a beam aberrated by a primary aberration with and without balancing as a function of its standard deviation σ_w in units of wavelength λ. (a) Uniform beam ($\gamma=0$) with a primary aberration. (b) Uniform beam with a balanced primary aberration. (c) Gaussian beam with $\gamma=1$ and a primary aberration. (d) Gaussian beam with $\gamma=1$ and a balanced primary aberration.

focus tolerance increases as γ increases. For a certain Strehl ratio, the value of B_d can be obtained from this figure, and the corresponding depth of focus can be obtained from Eq. (21). For example, a Strehl ratio of 0.8 is obtained for $B_d=\lambda/4$ when $\gamma\lesssim 1$. Hence the depth of focus is approximately $\pm 2\lambda F^2$. However, its corresponding value when $\sqrt{\gamma}=3$ is $\pm 5\lambda F^2$.

Figure 3 shows how the Strehl ratio varies with spherical aberration A_s with and without balancing with defocus. We note from Figs. 3(a) and 3(b) that the balanced aberration yields a much higher Strehl ratio for small values of γ but $A_s\lesssim 3\lambda$. Thus when spherical aberration $A_s\rho^4$ is present in a system, the central irradiance in a defocused observation plane yielding minimum aberration variance is higher than that at the Gaussian image point or the beam focus. For a weakly truncated Gaussian beam, however, the improvement in Strehl ratio due to balancing is not as significant, as illustrated in Fig. 3(c) for $\sqrt{\gamma}=3$. Moreover, in this case, the balancing of spherical aberration reduces the Strehl ratio when $A_s\gtrsim 8.3\lambda$. Of course, the value of B_d defining the defocused plane for minimum aberration variance depends on the value of γ.

Although for moderate values of A_s, the Strehl ratio for balanced spherical aberration is greater than its corresponding value without balancing with defocus, a larger value is obtained in a plane that is different from the one that yields minimum variance when the aberration is large. Figure 4(a) shows how the Strehl ratio of a uniform beam for various values of balanced spherical aberration varies as a function of a defocus aberration δ beyond that required for minimum variance of spherical aberration. We note that the Strehl ratio is symmetrical about $\delta=0$, as expected.[2,21] However, when $A_s\geqslant 2.3\lambda$, minimum variance does not yield maximum Strehl ratio. For example, when $A_s=3\lambda$, the maximum central irradiance is obtained in a plane corresponding to $\delta=\pm 1\lambda$, or $B_d=-2\lambda$ or -4λ. In fact, a minimum of irradiance is obtained in the plane $\delta=0$, or $B_d=-3\lambda$, corresponding to minimum variance. A similar result is obtained for a Gaussian beam with $\gamma=1$, as illustrated in Fig. 4(b), except that the Strehl ratio is not symmetric about $\delta=0$. For a weakly truncated Gaussian beam with $\sqrt{\gamma}=3$, maximum Strehl ratio is obtained for $\delta>0$ even for $A_s=2\lambda$, as shown in Fig. 4(c). The peak shifts to larger values of δ as A_s increases. As expected

from Eq. (22), the aberration-free axial irradiance is symmetric about the focal point regardless of the value of γ.

For a uniform pupil, balanced coma yields a higher Strehl ratio than the Seidel coma for $A_c \leq 2.3\lambda$, as illustrated in Fig. 5(a). For larger aberrations, Seidel coma yields a higher Strehl ratio than balanced coma, i.e., the irradiance at the focal point is higher than at the point with respect to which the aberration variance is minimum. Moreover, peak irradiance occurs at the point with respect to which the aberration variance is minimum for $A_c \leq 0.7\lambda$.[2,22] Similar results are obtained for a Gaussian beam, as illustrated in Figs. 5(b) and 5(c), although the crossover point occurs for larger values of A_c as γ increases. For $\sqrt{\gamma}=3$, for example, the crossover occurs at $A_c=7\lambda$. For larger aberrations, the difference between the Strehl ratios for balanced coma and Seidel coma increases as γ increases.

Figure 6 shows the Strehl ratio for astigmatism. The increase in Strehl ratio obtained by balancing astigmatism with defocus, like the decrease in σ_Φ, is not as significant as in the case of balanced spherical aberration or balanced coma. As in the case of coma, balanced astigmatism yields a higher Strehl ratio than Seidel astigmatism for $A_a \leq 1\lambda$ with small values of γ. For a weakly truncated Gaussian beam with $\sqrt{\gamma}=3$, the crossover point is $A_a \simeq 4\lambda$.

Since for small aberrations, the Strehl ratio depends on the aberration variance regardless of the type of aberration, it is useful to determine the range of validity of the approximate Eqs. (10). Figure 7 shows how the Strehl ratio varies with the standard deviation σ_w of the wave aberration in units of wavelength λ, where $\sigma_w=(\lambda/2\pi)\sigma_\Phi$. For a uniform pupil ($\gamma=0$), S_3, i.e., $\exp(-\sigma_\Phi^2)$, gives a reasonable approximation of Strehl ratio for $\sigma_w \leq \lambda/5$ for a primary aberration, as illustrated in Fig. 7(a). As shown in Fig. 7(b), this approximation is even better for a balanced primary aberration and up to $\sigma_w \leq \lambda/4$. As stated earlier, the Strehl ratio for given values of σ_w is the same for both spherical and balanced spherical aberrations.[2,8] S_3 yields the Strehl ratio within 10% of its true value as long it is ≥ 0.3.[2,8] Figures 7(c) and 7(d) show that the range of σ_w for which S_3 gives a reasonable approximation decreases somewhat as γ increases to unity. For $\sqrt{\gamma}=2$, the range decreases even more, as illustrated in Fig. 8(a). Moreover, except for $\sigma_w \leq \lambda/10$, the Strehl ratios for a given value of σ_w of the various aberrations are quite different from each other. In particular, the Strehl ratio for a balanced aberration is significantly smaller than its corresponding value without balancing. For a weakly truncated beam with $\sqrt{\gamma}=3$, Fig. 8(b) shows that, unless the Strehl ratio is ≥ 0.9 or $\sigma_w \leq \lambda/20$, S_3 significantly underestimates the true value of the Strehl ratio. Of course the approximate expressions S_1 and S_2 underestimate the Strehl ratio even more. The aberration coefficient A_s is now quite large compared with its corresponding σ_w, which is different for the aberration with and without balancing. From Tables 1 and 3, $\sigma_w=\lambda/4$ for a weakly truncated Gaussian beam with $\sqrt{\gamma}=3$ corresponds to $A_s=4.5\lambda$ and 10.1λ of spherical and balanced spherical aberration, $A_c=3.9\lambda$ and 6.8λ of coma and balanced coma, and $A_a=3.2\lambda$ and 4.5λ of astigmatism and balanced astigmatism.

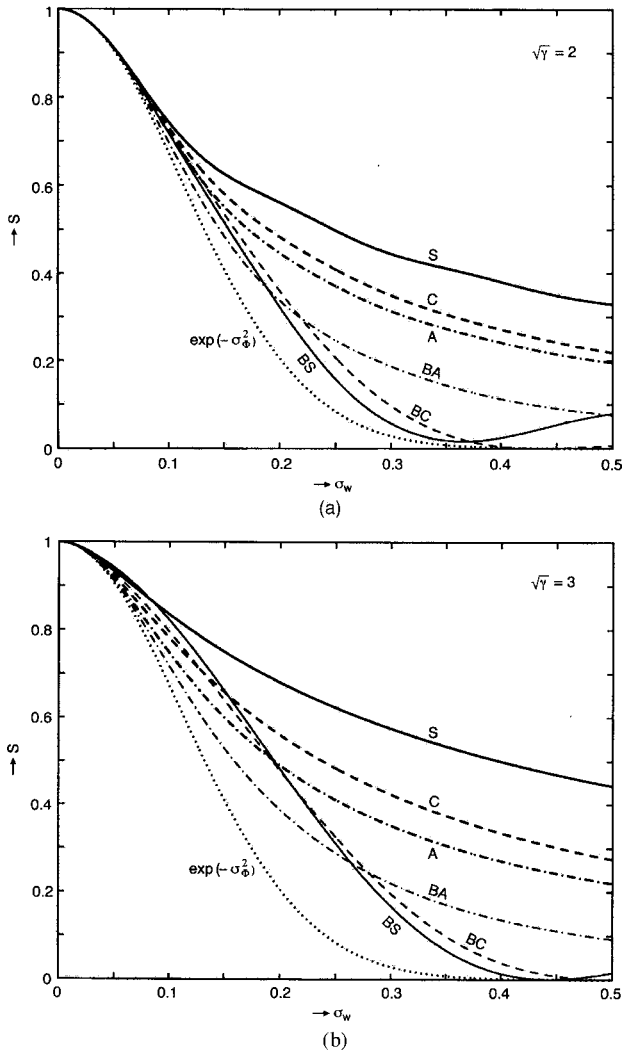


Fig. 8. Strehl ratio of a Gaussian beam aberrated by a primary aberration compared with its value for a corresponding balanced primary aberration as a function of standard deviation σ_w in units of wavelength λ. S, spherical; BS, balanced spherical; C, coma; BC, balanced coma; A, astigmatism; and BA, balanced astigmatism. (a) $\sqrt{\gamma}=2$. (b) Weakly truncated beam with $\sqrt{\gamma}=3$.

In view of these large aberrations, validity of the aberration variance as a criterion for the Strehl ratio may be in question and the results of Fig. 8 perhaps not surprising. In particular, the portions of the wavefront with integral and near integral number of wavelengths of aberration contribute to the Strehl ratio in and nearly in phase, thereby increasing its value. However, σ_Φ increases linearly with an aberration coefficient A_i, and S_3 yields smaller and smaller values for the Strehl ratio.

8. CONCLUSIONS

For small aberrations, the balancing of an aberration to reduce its variance σ_Φ^2 improves the Strehl ratio. The amount of a primary aberration for which this is true increases with the truncation parameter $\gamma=(a/\omega)^2$. For $\gamma \leq 1$, $\exp(-\sigma_\Phi^2)$ yields a good estimate of the true value of the Strehl ratio. As γ increases, the range of σ_w for which

it gives a reasonable estimate of the Strehl ratio decreases. For a weakly truncated beam, it gives a much smaller value than the true value unless the Strehl ratio is $\gtrsim 0.9$ or $\sigma_w \lesssim \lambda/20$.

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REFERENCES

1. K. Strehl, "Über Luftschlieren und Zonenfehler," Z. Instrumentenkd. **22**, 213–217 (1902).
2. V. N. Mahajan, *Optical Imaging and Aberrations, Part II: Wave Diffraction Optics*, 2nd printing (SPIE Press, 2004).
3. M. Born and E. Wolf, *Principles of Optics*, 7th ed. (Oxford, New York, 1999).
4. V. N. Mahajan, "Strehl ratio for primary aberrations: some analytical results for circular and annular pupils," J. Opt. Soc. Am. **72**, 1258–1266 (1982).
5. V. N. Mahajan, "Strehl ratio for primary aberrations: some analytical results for circular and annular pupils: errata," J. Opt. Soc. Am. A **10**, 2092 (1993).
6. D. D. Lowenthal, "Maréchal intensity criteria modified for Gaussian beams," Appl. Opt. **13**, 2126–2133 (1974).
7. D. D. Lowenthal, "Maréchal intensity criteria modified for Gaussian beams: errata," Appl. Opt. **13**, 2774 (1974).
8. V. N. Mahajan, "Strehl ratio for primary aberrations in terms of their aberration variance," J. Opt. Soc. Am. **73**, 860–861 (1983).
9. R. Herloski, "Strehl ratio for untruncated Gaussian beams," J. Opt. Soc. Am. A **2**, 1027–1030 (1985).
10. V. N. Mahajan, "Uniform versus Gaussian beams: a comparison of the effects of diffraction, obscuration, and aberrations," J. Opt. Soc. Am. A **3**, 470–485 (1986).
11. A. E. Siegman, *An Introduction to Lasers and Masers* (McGraw Hill, 1971).
12. J. D. Gaskill, *Linear Systems, Fourier Transforms, and Optics* (McGraw Hill, 1978).
13. V. N. Mahajan, "Axial irradiance of a focused beam," J. Opt. Soc. Am. A **22**, 1814–1823 (2005).
14. S. Szapiel, "Aberration balancing techniques for radially symmetric amplitude distributions; a generalization of the Maréchal approach," J. Opt. Soc. Am. **72**, 947–956 (1982).
15. V. N. Mahajan, "Zernike annular polynomials for imaging systems with annular pupils," J. Opt. Soc. Am. **71**, 75–85 (1981).
16. V. N. Mahajan, "Zernike annular polynomials for imaging systems with annular pupils," Errata, J. Opt. Soc. Am. **71**, 1408 (1981).
17. V. N. Mahajan, "Zernike annular polynomials for imaging systems with annular pupils," JOSA Communication, J. Opt. Soc. Am. A **1**, 685 (1984).
18. V. N. Mahajan, "Zernike-Gauss polynomials for optical systems with Gaussian pupils," Appl. Opt. **34**, 8057–8059 (1995).
19. V. N. Mahajan, "Zernike polynomials and aberration balancing," Proc. SPIE **5173**, 1–7 (2003).
20. G. Martial, "Strehl ratio and aberration balancing," J. Opt. Soc. Am. A **8**, 164–170 (1991).
21. V. N. Mahajan, "Symmetry properties of aberrated point-spread functions," J. Opt. Soc. Am. A **11**, 1993–2003 (1994).
22. V. N. Mahajan, "Line of sight of an aberrated optical system," J. Opt. Soc. Am. A **2**, 833–846 (1985).